AUTODOC AI

Sample Automotive High-Level Design (HLD)

System	Obstacle Detection and Automatic Braking Support System
Document ID	SYN-HLD-ADAS-001
Version	1.0 — synthetic sample
Purpose	Test document ingestion, page-aware retrieval, and evidence-grounded Q&A in AutoDoc AI.

Important notice

This document is entirely synthetic and was created for educational software testing. It is not a real Tata Technologies, OEM, AUTOSAR, or production-vehicle design document. The component names and behaviours are illustrative only.

How to use this PDF

1. Open AutoDoc AI in your browser.
2. Upload this PDF using the upload control.
3. Check whether page counts and extracted text appear correctly.
4. Later, use the sample questions in your evaluation set to test retrieval and citations.

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1. Purpose and Scope

The illustrative Obstacle Detection and Automatic Braking Support System demonstrates how an automotive system can be described at a high level. A forward-facing obstacle sensor sends detection information to a Driver Assistance Controller. The controller evaluates the input and, when the configured conditions are met, sends a braking request to a Brake Control Interface.

Scope of this sample

Included: component responsibilities, example interfaces, a simplified data flow, and sample requirements for testing document question answering. Excluded: real vehicle control logic, calibrated thresholds, production safety analysis, certified requirements, and implementation-ready braking algorithms.

2. System Context

Component	Input	Output	Responsibility
Obstacle Sensor	Reflected signal / sensor observations	ObstacleDetectionMessage	Produces an illustrative obstacle detection message.
Driver Assistance Controller	ObstacleDetectionMessage	BrakeRequestMessage or NoRequest	Evaluates the received detection message against configured example conditions.
Brake Control Interface	BrakeRequestMessage	BrakeCommandStatus	Forwards an approved request to a downstream braking subsystem in a real implementation; this sample does not implement vehicle control.
Driver Display	SystemStatusMessage	WarningIndicator	Displays a simplified system status or warning.

3. High-Level Data Flow

1. Obstacle Sensor	→	2. Driver Assistance Controller	→	3. Brake Control Interface
Produces detection message		Evaluates message and configured conditions		Receives a brake request

4. Interfaces and Example Data

Interface / message	Producer → Consumer	Example fields
ObstacleDetectionMessage	Obstacle Sensor → Driver Assistance Controller	timestamp, detection_present, estimated_distance_m, confidence
BrakeRequestMessage	Driver Assistance Controller → Brake Control Interface	request_id, request_state, reason_code
BrakeCommandStatus	Brake Control Interface → Driver Assistance Controller	request_id, status, status_timestamp
SystemStatusMessage	Controller → Driver Display	system_state, warning_code

5. Example Requirements and Limitations

• **REQ-01:** The controller should check that an obstacle detection message contains the fields required by this sample design before evaluating it.

• **REQ-02:** A brake request should include a request identifier and a reason code.

• **REQ-03:** The driver display should present a warning when the system status indicates a warning condition.

• **REQ-04:** The application analysing this document should cite the source page when answering questions about these components and interfaces.

Limitations: These are synthetic example requirements, not validated automotive safety requirements. No real braking thresholds, safety integrity levels, AUTOSAR configuration, or vehicle-specific protocols are specified. Engineering review is mandatory before any real-world use.

6. Sample Evaluation Questions

Question	Expected answer from this sample	Source page
Which component produces ObstacleDetectionMessage?	Obstacle Sensor.	2–3
Which component evaluates the obstacle detection message?	Driver Assistance Controller.	2
Which interface receives BrakeRequestMessage?	Brake Control Interface.	2–3
Name two fields in ObstacleDetectionMessage.	Examples: timestamp, detection_present, estimated_distance_m, confidence.	3
Does this document specify a real braking distance threshold?	No. It explicitly states that real braking thresholds are not specified.	2–3
Which component displays the system warning?	Driver Display.	2–3

Evaluation note: Use these questions as a starting ground-truth set. Verify actual retrieval and citation results in AutoDoc AI and report the measured results; do not assume the system answered correctly without testing.